Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🏴 Flag 0 — Create Your Feature Branch

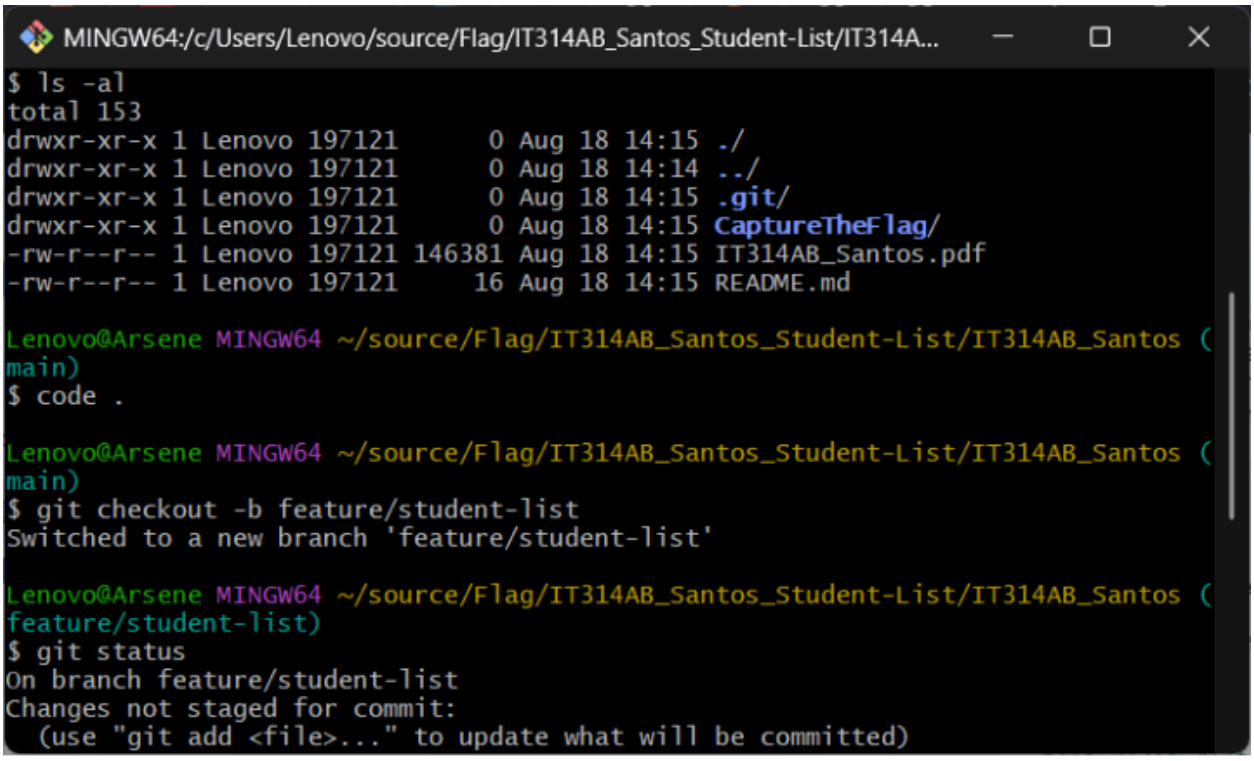
Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

- Terminal showing your current branch



🏴 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart `List`.

Research what a Dart `List` is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

- Image
- Name
- Course
- Year Level
- Age
- Hobby

Your data should exist separately from your UI widgets.

Screenshot:

Code:

- Your student list
- At least five student entries

```dart
const List<Profile> profiles = [
  Profile(
    name: 'Name: Carl Joshua Santos',
    hobby: 'Hobby: Sleeping',
    height: 5.8399,
    isStudent: true,
    color: 0xFF1E3A5F,
    picture: "assets/shh.jpg",
  ),

  Profile(
    name: 'Name: Anthony Josh Moreno',
    courseAndSection: 'Course And Section: BSIT-3',
    age: 300,
    height: 5.4,
    isStudent: true,
    color: 0xFF155E63,
    picture: "assets/tonikss.jpg",
  ),

  Profile(
    name: 'Name: Armanes Christian Artajo',
    hobby: 'Hobby: Playing Hearts',
    height: 5.8,
    isStudent: true,
    color: 0xFF245B45,
    picture: "assets/manes.jpg",
  ),

  Profile(
    name: 'Name: Krylle Jane Racaza',
    courseAndSection: 'Course And Section: BSIT-3C',
    hobby: 'Hobby: Sleeping',
    height: 5.2,
    isStudent: true,
    color: 0xFF7A263A,
    picture: "assets/karil.jpg",
  ),

  Profile(
    name: 'Name: Krylle Jane Racaza',
    courseAndSection: 'Course And Section: BSIT-3C',
    hobby: 'Hobby: Sleeping',
    height: 5.2,
    isStudent: true,
    color: 0xFF7A263A,
    picture: "assets/karil.jpg",
  ),

  Profile(
    name: 'Name: Angeline Ebarola',
    courseAndSection: 'Course And Section: BCE-3',
    age: 20,
    hobby: 'Hobby: Sleeping',
    height: 5.5,
    isStudent: true,
    color: 0xFF4A2C6D,
    picture: "assets/priiHAHAHAHA.jpg",
  ),
];
```

🏴 Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

Imagine having:

- 50 students.
- 500 students.
- 5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:

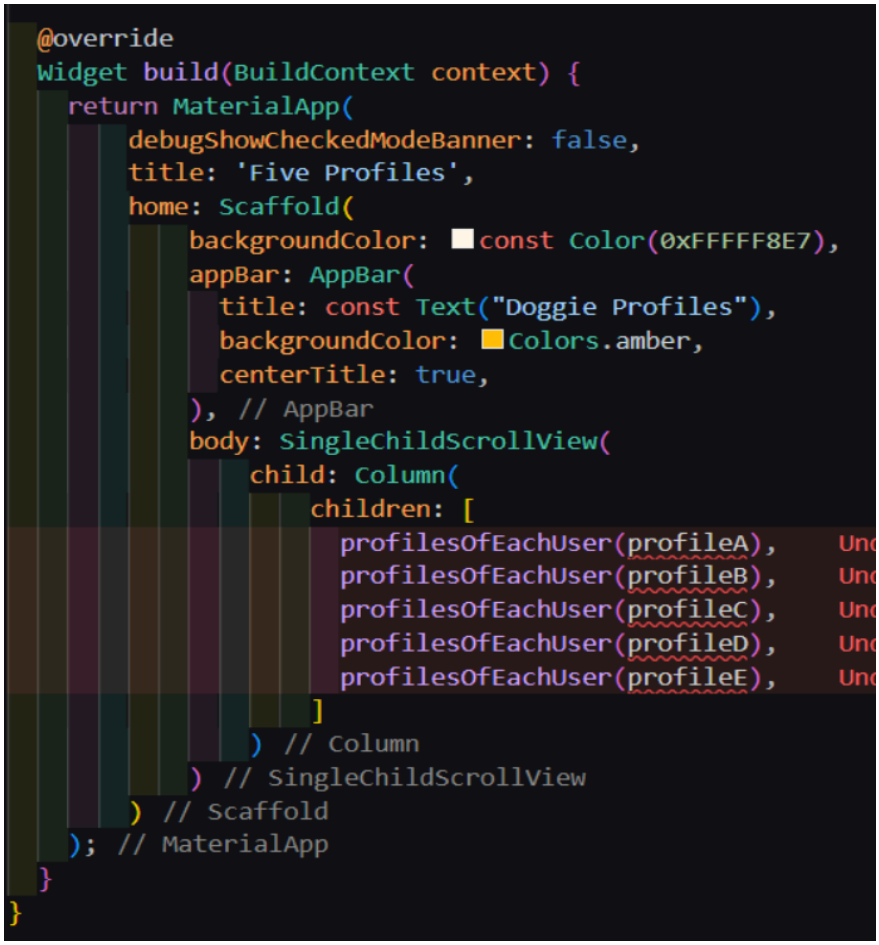
- Your five manually created student cards (do not change your past code yet)

```
class Profiles extends StatelessWidget {
  const Profiles({super.key});

  Widget profilesOfEachUser(Profile profile)
  {
    return SizedBox(
      width: 350,
      height: 400,
      child: Card(

        color: Color(profile.color),
        child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            crossAxisAlignment: CrossAxisAlignment.center,
            children: [
              CircleAvatar(
                radius: 55,
                backgroundImage: AssetImage(profile.picture),
              ), // CircleAvatar
              Text(profile.name ?? "Name Unknown",
                  style: const TextStyle(
                      color : Colors.white,
                      fontSize: 22,
                      fontWeight: FontWeight.bold) // TextStyle
              ), // Text
              Text(profile.courseAndSection ?? "Not Provided",style: TextStyle(color: Colors.white)),
              Text(profile.age?.toString() ?? "Not Provided",style: TextStyle(color: Colors.white)),
              Text(profile.hobby ?? "Not Provided",style: TextStyle(color: Colors.white)),
            ]
        ), // Column
      ), // Card
    ); // SizedBox
  }
```

// Nag method man ko ma’am, tas nag instance pag last code 😭



🏴 Flag 3 — Discover `ListView`

Objective: Make your student directory scrollable using `ListView`.

Research Flutter's `ListView`.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

Your application should now allow continuous vertical scrolling.

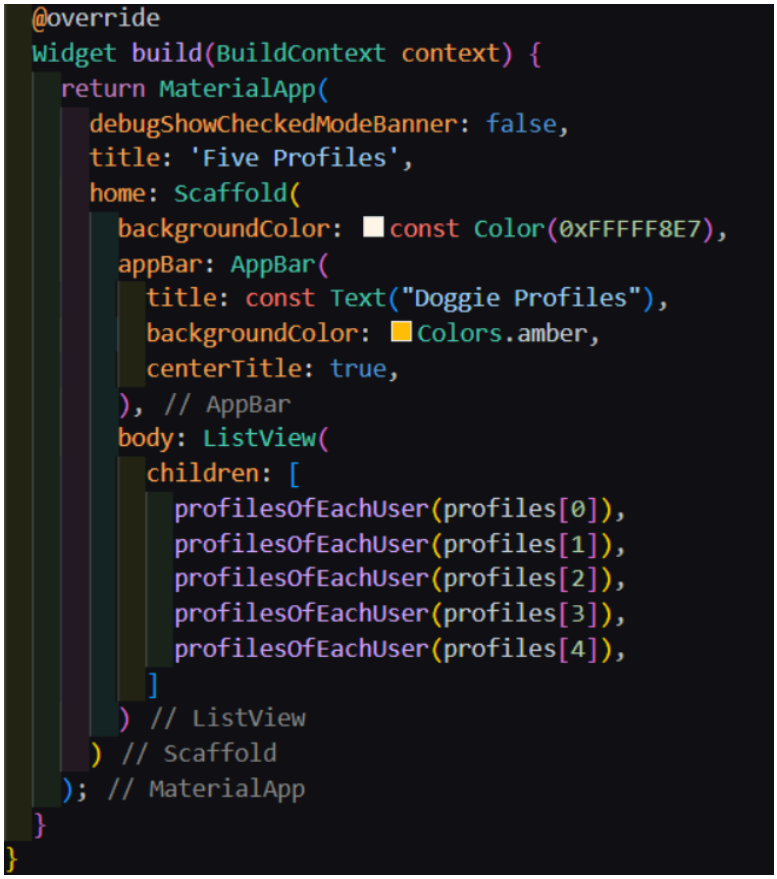
Test

The user should be able to:

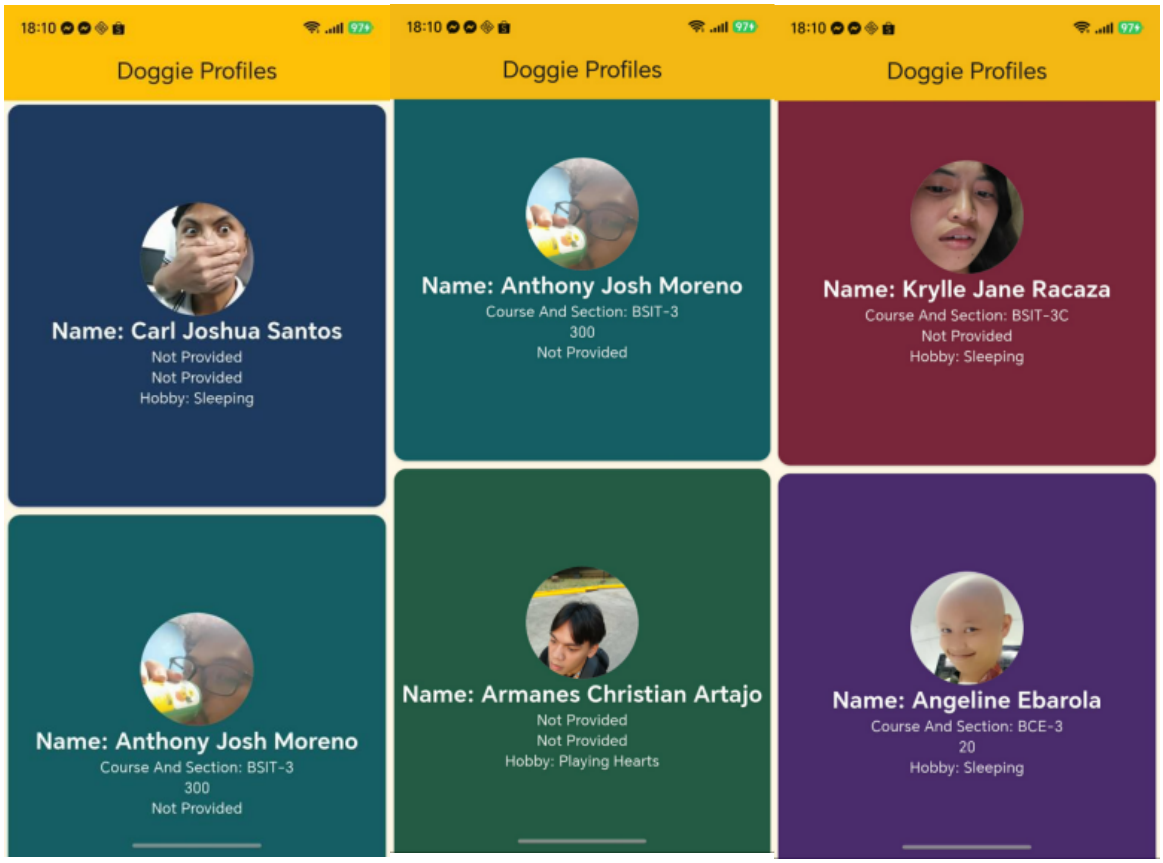
- Scroll from the first student
- Continue scrolling
- Reach the last student without resizing the screen

Screenshot:

- `ListView` in your code
- Top of the application
- Bottom portion showing later students



```
@override
Widget build(BuildContext context) {
  return MaterialApp(
    debugShowCheckedModeBanner: false,
    title: 'Five Profiles',
    home: Scaffold(
      backgroundColor: const Color(0xFFFFF8E7),
      appBar: AppBar(
        title: const Text("Doggie Profiles"),
        backgroundColor: Colors.amber,
        centerTitle: true,
      ), // AppBar
      body: ListView(
        children: [
          profilesOfEachUser(profiles[0]),
          profilesOfEachUser(profiles[1]),
          profilesOfEachUser(profiles[2]),
          profilesOfEachUser(profiles[3]),
          profilesOfEachUser(profiles[4]),
        ]
      ) // ListView
    ) // Scaffold
  ); // MaterialApp
}
}
```



🏴 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

Your UI should now create one card for each student in your list.

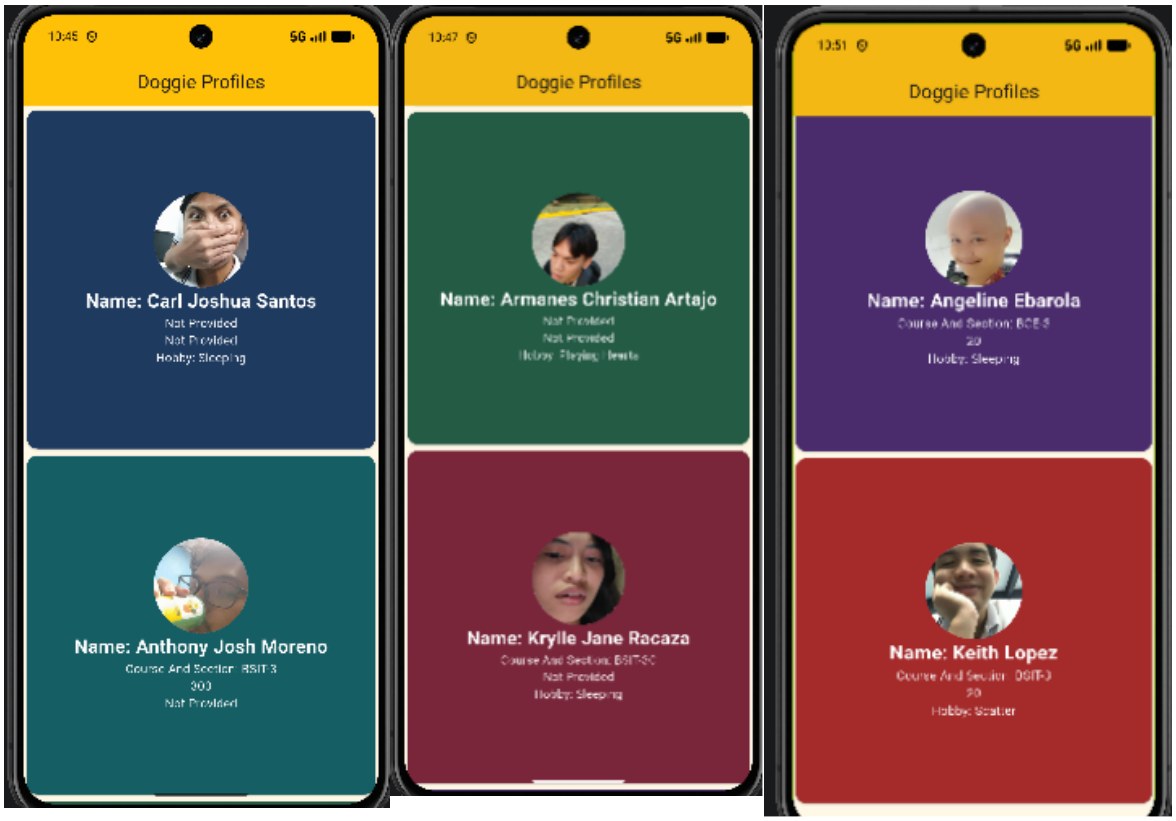
Test

Add one new student to your list.

Without creating another card manually, your application should display the new student automatically.

Screenshot:

- `ListView.builder` in your code
- Student list
- Newly generated student appearing in the app




```
Profile(
  name: 'Name: Keith Lopez',
  courseAndSection: 'Course And Section: BSIT-3',
  age: 20,
  hobby: 'Hobby: Scatter',
  height: 5.11,
  isStudent: true,
  color: 0xFFA52A2A,
  picture: "assets/Keith.jpg",
)
];
```

🏴 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

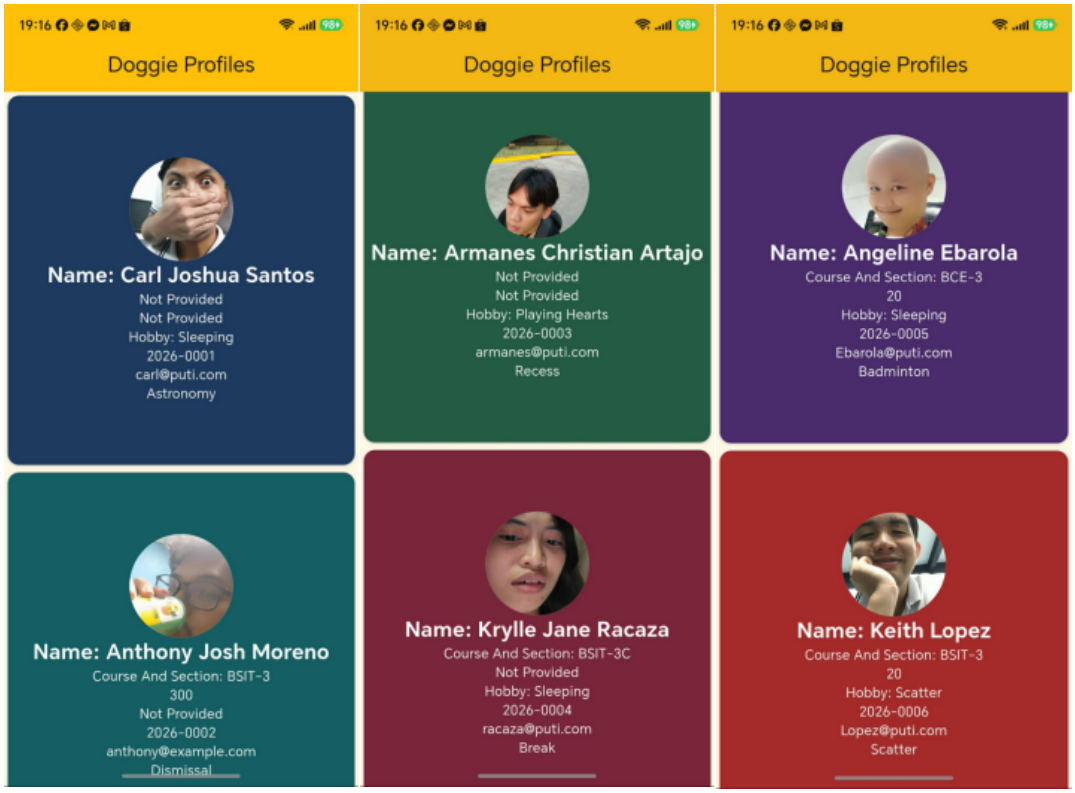
Examples:

- Student ID
- Email
- Favorite Subject

Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

- Updated student data
- Student Card displaying the additional fields



```dart
class Profile {
  final String? name;
  final String? courseAndSection;
  final int? age;
  final String? hobby;
  final double? height;
  final bool? isStudent;
  final int color;
  final String picture;
  final String? studentId;
  final String? email;
  final String? favoriteSubject;
  const Profile(
      { required this.name,  this.courseAndSection,  this.age,  this.hobby, required this.height,
      required this.isStudent, required this.color,required this.picture,    this.studentId,
        this.email,
        this.favoriteSubject,
      });
}

const List<Profile> profiles = [
  Profile(
    name: 'Name: Carl Joshua Santos',
    hobby: 'Hobby: Sleeping',
    height: 5.8399,
    isStudent: true,
    color: 0xFF1E3A5F,
    picture: "assets/shh.jpg",
    studentId: '2026-0001',
    email: 'carl@puti.com',
    favoriteSubject: 'Astronomy',
  ),
```

```dart
  Profile(
    name: 'Name: Anthony Josh Moreno',
    courseAndSection: 'Course And Section: BSIT-3',
    age: 300,
    height: 5.4,
    isStudent: true,
    color: 0xFF155E63,
    picture: "assets/tonikss.jpg",
    studentId: '2026-0002',
    email: 'anthony@example.com',
    favoriteSubject: 'Dismissal',
  ),

  Profile(
    name: 'Name: Armanes Christian Artajo',
    hobby: 'Hobby: Playing Hearts',
    height: 5.8,
    isStudent: true,
    color: 0xFF245B45,
    picture: "assets/manes.jpg",
    studentId: '2026-0003',
    email: 'armanes@puti.com',
    favoriteSubject: 'Recess',
  ),

  Profile(
    name: 'Name: Krylle Jane Racaza',
    courseAndSection: 'Course And Section: BSIT-3C',
    hobby: 'Hobby: Sleeping',
    height: 5.2,
    isStudent: true,
    color: 0xFF7A263A,
    picture: "assets/karil.jpg",
    studentId: '2026-0004',
    email: 'racaza@puti.com',
    favoriteSubject: 'Break',
```

```dart
  Profile(
    name: 'Name: Angeline Ebarola',
    courseAndSection: 'Course And Section: BCE-3',
    age: 20,
    hobby: 'Hobby: Sleeping',
    height: 5.5,
    isStudent: true,
    color: 0xFF4A2C6D,
    picture: "assets/priiHAHAHAHA.jpg",
    studentId: '2026-0005',
    email: 'Ebarola@puti.com',
    favoriteSubject: 'Badminton',
  ),

  Profile(
    name: 'Name: Keith Lopez',
    courseAndSection: 'Course And Section: BSIT-3',
    age: 20,
    hobby: 'Hobby: Scatter',
    height: 5.11,
    isStudent: true,
    color: 0xFFA52A2A,
    picture: "assets/Keith.jpg",
    studentId: '2026-0006',
    email: 'Lopez@puti.com',
    favoriteSubject: 'Scatter',
  )
];
```

```
Text(profile.courseAndSection ?? "Not Provided",style: TextStyle(color: Colors.white)),
Text(profile.age?.toString() ?? "Not Provided",style: TextStyle(color: Colors.white)),
Text(profile.hobby ?? "Not Provided",style: TextStyle(color: Colors.white)),
Text(profile.studentId ?? "Not Provided",style: TextStyle(color: Colors.white)),
Text(profile.email ?? "Not Provided",style: TextStyle(color: Colors.white)),
Text(profile.favoriteSubject ?? "Not Provided",style: TextStyle(color: Colors.white)),
```

```
@override
Widget build(BuildContext context) {
  return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Five Profiles',
      home: Scaffold(
          backgroundColor: const Color(0xFFFFF8E7),
          appBar: AppBar(
            title: const Text("Doggie Profiles"),
            backgroundColor: Colors.amber,
            centerTitle: true,
          ), // AppBar
          body: ListView.builder(
            itemCount: profiles.length,
            itemBuilder: (context, index) {
              return profilesOfEachUser(profiles[index]);
            },
          ) // ListView.builder
      ) // Scaffold
  ); // MaterialApp
  }
}
```

🏴 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

[]

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

Examples:

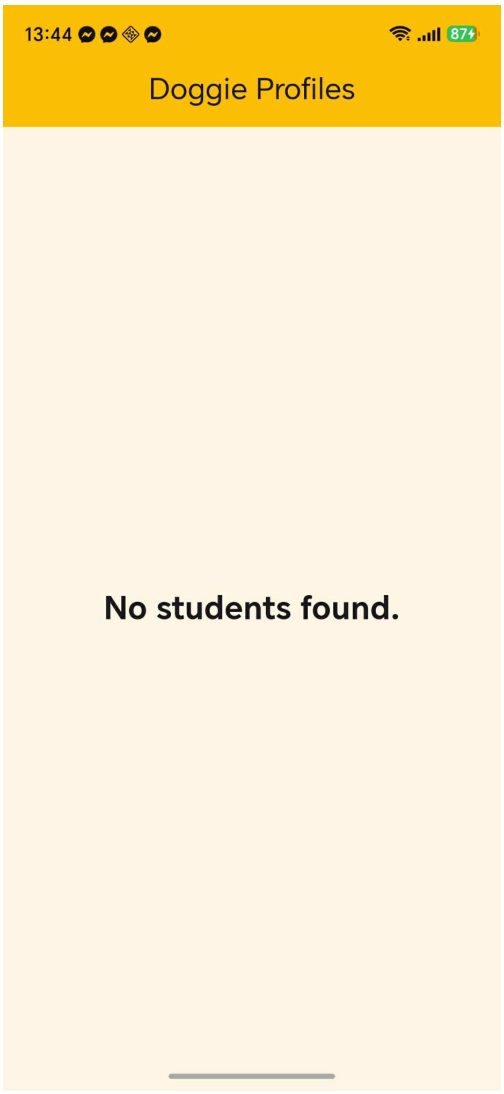
- "No students found."
- "Student list is empty."

Test

- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:

- Empty student list
- Empty-state UI
- Application still functioning correctly



🏴 Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a `List`.

Sort your students alphabetically by name.

Test

Before sorting:

- Maria

- John
- Aaron
- Sophia

After sorting:

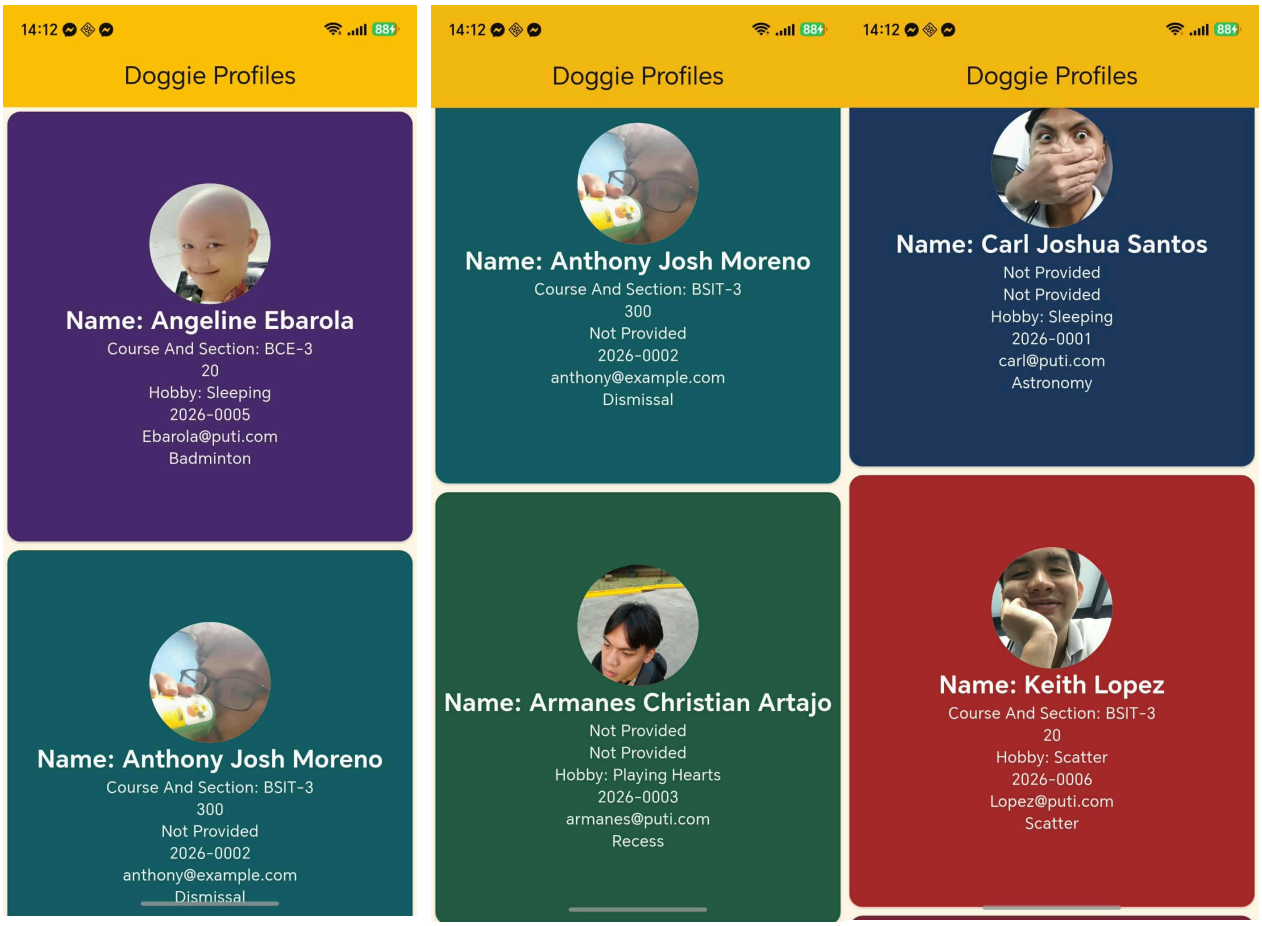
- Aaron
- John
- Maria
- Sophia

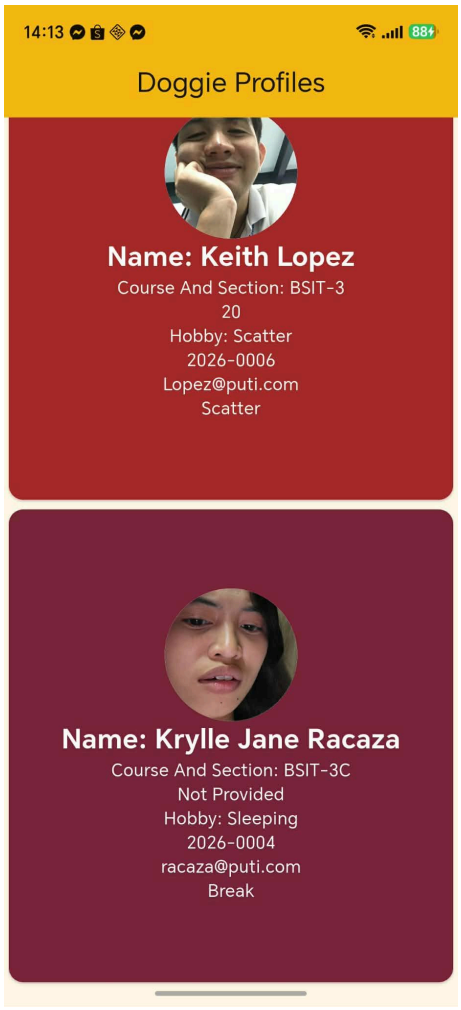
Your UI should immediately reflect the new order.

Screenshot:

- Student list before or after sorting
- Application displaying the sorted order

```dart
@override
Widget build(BuildContext context) {
  profiles.sort((a,b) => (a.name ?? "").compareTo(b.name ?? ""));
  return MaterialApp(
```





🏴 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

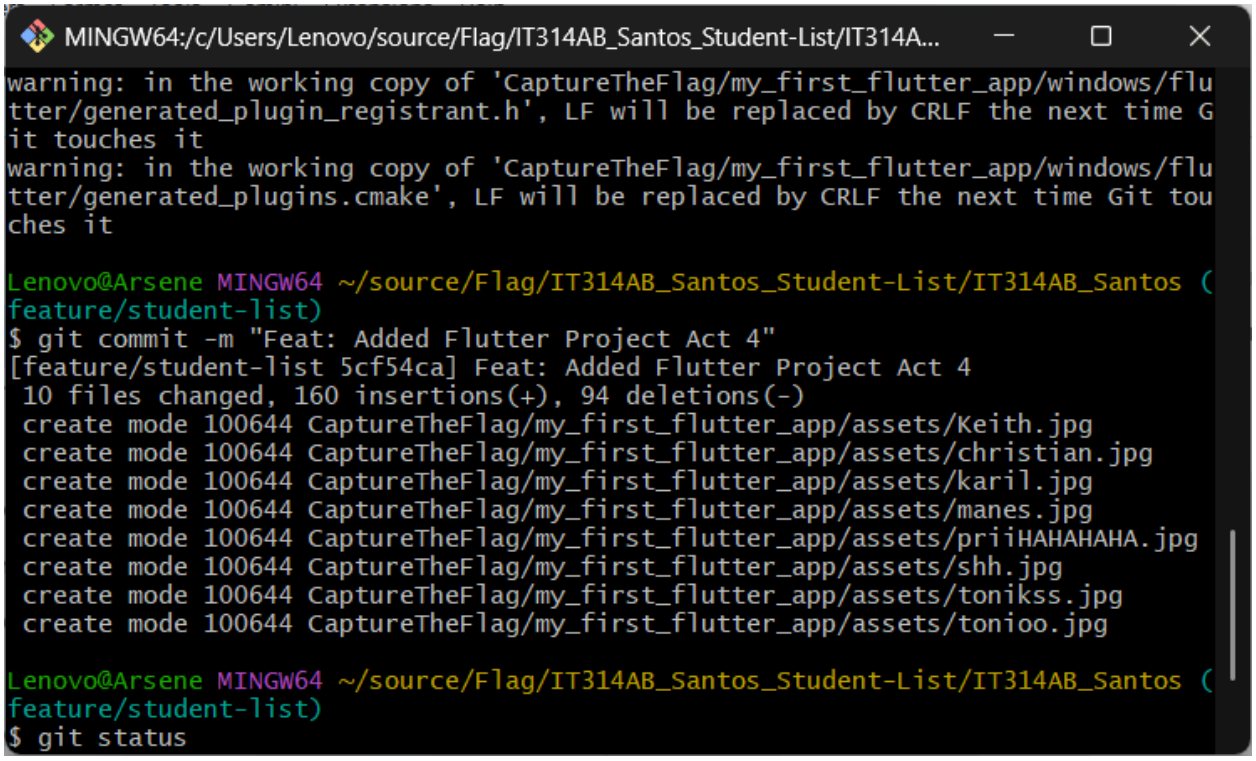
Use an appropriate commit message following proper Git standards.

Branch:

`feature/student-list`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit


MINGW64:/c/Users/Lenovo/source/Flag/IT314AB_Santos_Student-List/IT314A...
warning: in the working copy of 'CaptureTheFlag/my_first_flutter_app/windows/flutter/generated_plugin_registrant.h', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'CaptureTheFlag/my_first_flutter_app/windows/flutter/generated_plugins.cmake', LF will be replaced by CRLF the next time Git touches it
Lenovo@Arsene MINGW64 ~/source/Flag/IT314AB_Santos_Student-List/IT314AB_Santos (feature/student-list)
$ git commit -m "Feat: Added Flutter Project Act 4"
[feature/student-list 5cf54ca] Feat: Added Flutter Project Act 4
10 files changed, 160 insertions(+), 94 deletions(-)
create mode 100644 CaptureTheFlag/my_first_flutter_app/assets/Keith.jpg
create mode 100644 CaptureTheFlag/my_first_flutter_app/assets/christian.jpg
create mode 100644 CaptureTheFlag/my_first_flutter_app/assets/karil.jpg
create mode 100644 CaptureTheFlag/my_first_flutter_app/assets/manes.jpg
create mode 100644 CaptureTheFlag/my_first_flutter_app/assets/priiHAHAHAHA.jpg
create mode 100644 CaptureTheFlag/my_first_flutter_app/assets/shh.jpg
create mode 100644 CaptureTheFlag/my_first_flutter_app/assets/tonikss.jpg
create mode 100644 CaptureTheFlag/my_first_flutter_app/assets/tonioo.jpg
Lenovo@Arsene MINGW64 ~/source/Flag/IT314AB_Santos_Student-List/IT314AB_Santos (feature/student-list)
$ git status